



County Oberspreewald-Lausitz

May 2022

Citizen information

From lignite tradition to modern technologies - Lusatia becomes an important player in the traffic and transport revolution



Dekra Lausitzring
Lausitzallee 1
01998 Klettwitz

<https://www.dekra-lausitzring.de/>



Tesla Motors, Inc.
Gigafactory Berlin-Brandenburg
Tesla Straße 1
15537 Grünheide
Germany

<https://www.tesla.com/>



Federal State Brandenburg
Ministries for Economy, Work and Energy
Agriculture, Environment and Climate Protection
Science, Research and Arts
State Office
14473 Potsdam

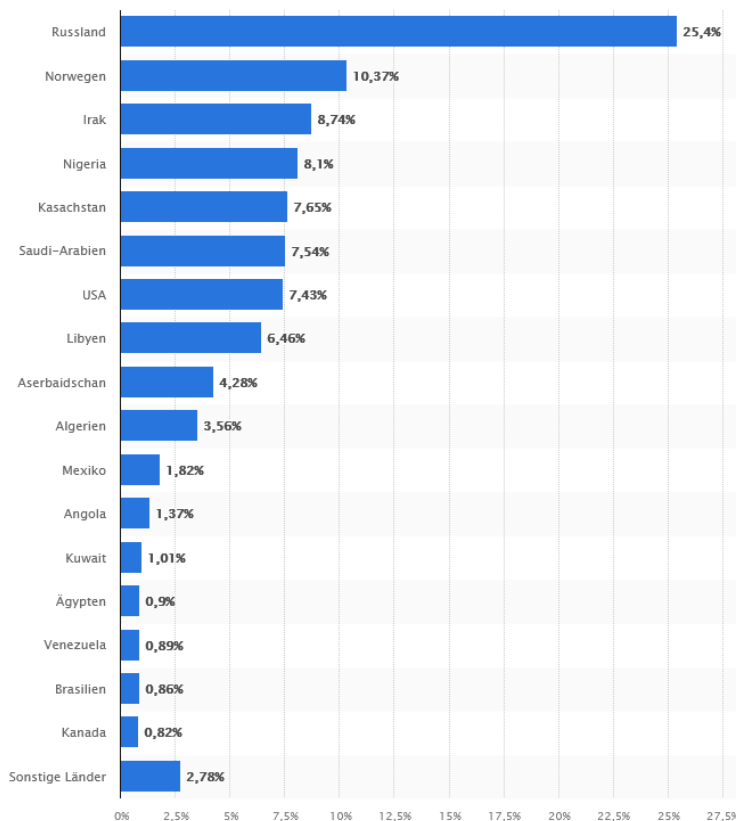
<https://www.brandenburg.de/>

The General Background

A far-reaching rethink is needed. This can certainly be stated as an economic mantra for the next two decades. The important terms that are shaping the political discussion today in 2022 are:

- Energy transition and
- transport turnaround

The two are directly related. Not least the war in Ukraine has made it clear to the industrially developed countries of the European Union, Great Britain as well as the USA, Canada and Japan that they will need new answers for the raw materials issues in the coming years. As German Finance Minister Christian Lindner said in a speech on Feb. 27, 2022, at the start of the Ukraine war, "Renewable energies are freedom energies." In recent weeks, the Ukraine war revealed the great dependence of Germany but also of the EU as a whole on imports of fossil fuels. However, the political discussion about the attempts to find substitute suppliers in the short term also shows that many of the major exporters of fossil fuels belong to countries whose political systems only partially correspond to our Western ideas of liberal democracies. The following overview shows the distribution of the EU's oil imports in 2019.



Quelle Statistisches Bundesamt (2020):
<https://de.statista.com/statistik/daten/studie/172674/umfrage/verteilung-der-oelimporte-der-eu-nach-herkunft/>

In the field of traffic and transport, however, crude oil is only one raw material. It dominates the discussion, especially in the current situation, as it is closely linked in the public's consciousness with CO2 emissions and thus with climate change. This is a particular threat for Brandenburg. The state, which is already very dry, is severely affected by rising temperatures and changes in the frequency and amount of precipitation. Especially in the heat summers, the risk of forest fires rises sharply. For March 2022, a fivefold increase in the number of forest fires compared to last year has already been reported.

Against the backdrop of the Ukraine war and the sanctions against Russia, China's political ambitions with regard to Taiwan are also playing an increasing role. Mainly due to Corona, we can currently experience how sanctions against China can af-

fect the supply chains of European industry. Automotive delivery times of 12 months and the discovery that products are not available for delivery show our dependence on a global superpower that also follows a very different set of values with regard to freedom values.

Traffic and Transport for a livable future

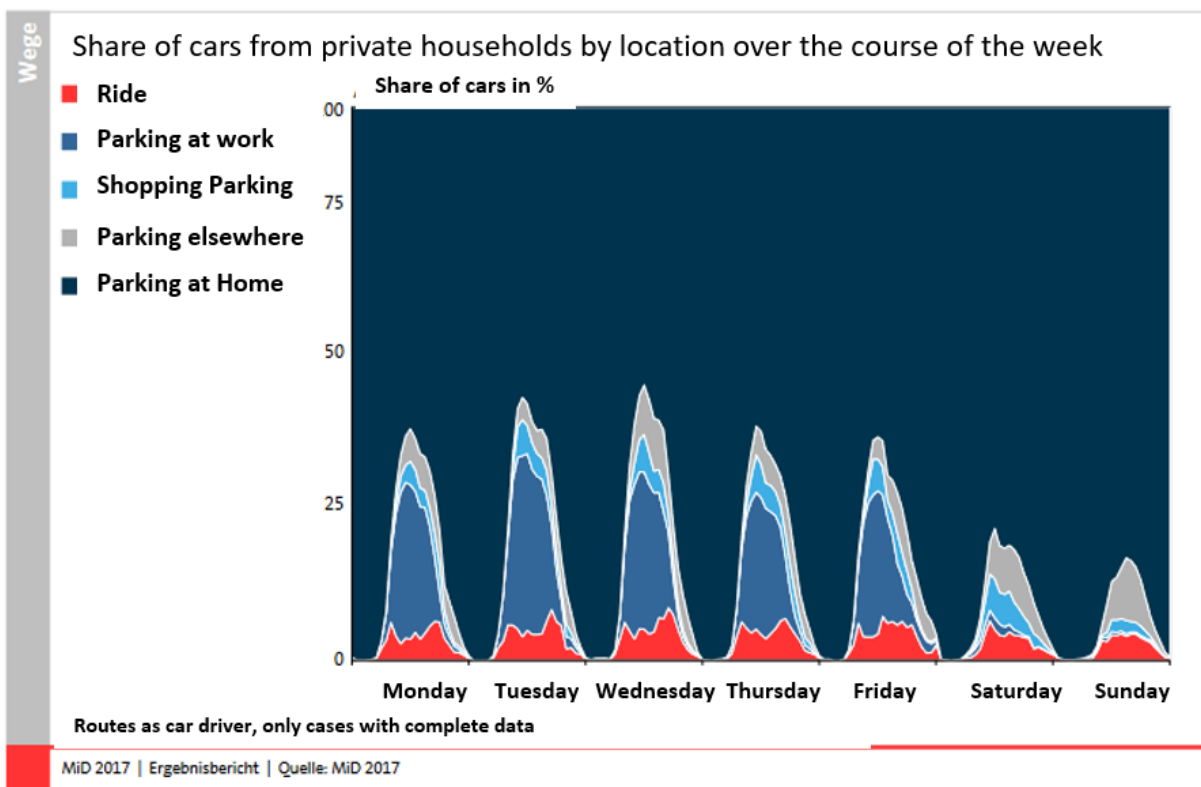
Particularly in the area of traffic and transport, we all feel that changes are needed that will have to be implemented in the coming decades. In the years of the economic miracle and the economic upswing, the car was a promise of freedom for the individual. Getting from A to B quickly and flexibly at any time was part of our social concept and shaped urban planning, spatial development, economic policy, but also the everyday behavior and habits of each individual. From the perspective of society as a whole, individual freedom is offset by the consumption of land and resources, which will also have a negative impact on the living conditions of future generations. The following two illustrations, which are taken from the last panel survey "Mobility in Germany" (p. 76 and p. 77), show the abundance in which large parts of the German population live.

Both graphs show how little time cars are used during the course of the day or week, and to what extent the vehicles are used during the week. In particular, the graph shows that more than 50% of cars are always parked at home (i.e. not used) during the entire week. In particular, the red area of the graph also shows that even during rush hour only about 10% of the available cars are used.

Table 11 Average travel times and idle times by location per car and day

<u>Routes as car driver</u> (only cases with complete data)	<u>h:min</u>	<u>Hours per day</u>
Ride	h:min	00:46
Parking at work	h:min	01:42
Shopping Parking	h:min	00:27
Parking at home	h:min	20:15
Parking elsewhere	h:min	00:50
Total	h:min	24:00

When we look at current transportation and established travel behavior, we see the need for major changes. What could traffic and transport of the future look like that better and consistently takes into account the needs of future generations and thus sustainability, but at the same time currently guarantees the mobility options that exist today?

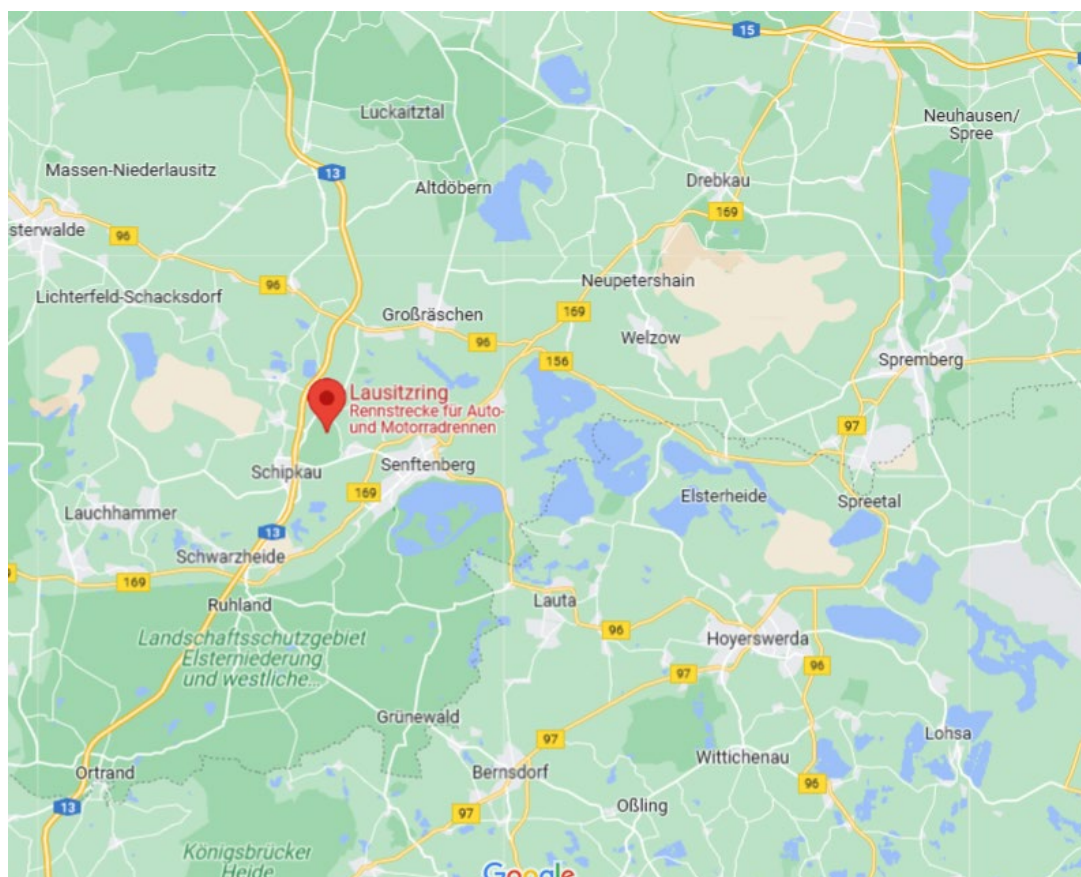


Initiative of the

District of Oberspreewald-Lausitz, DEKRA Lausitzring and Tesla Inc.
(with the support of the Federal State of Brandenburg)

We have joined forces as an initiative to develop the scientific basis for a sustainable mobility concept. With the location of Tesla Inc. in Grünheide, a close cooperation with the state of Brandenburg has developed. This has now been extended to include two of the above-mentioned partners. The aim of the joint initiative is to support the research and development efforts in the field of autonomous driving that Tesla Inc. has already been pursuing for some time. Talks between the operators of the Dekra Lausitzring and Tesla Inc. about using the Lausitzring for test drives began about a year ago. In particular, the possibilities of autonomous driving are to be improved and tested in many different traffic situations on a laboratory scale at the Lausitzring. The goal is to advance the development of autonomous driving to such an extent that, in the medium term, the fourth level is reached via the third level, in which the driver becomes a passenger and completely relinquishes control of the vehicle. The vehicle will then cover distances independently, and all occupants will be able to sleep or read the newspaper or even work. In addition, these autonomous vehicles can then also drive without passengers and thus create new possibilities for changing traffic and transport in rural areas (a brief overview of the stages/levels of autonomous driving can be found here <https://www.apiv.com/en/insights/article/what-are-the-levels-of-automated-driving>)

The map section on the following page shows the location and the surrounding region of about 50 x 50 km. At about 130 km, the Lausitzring is not far from the Tesla Gigafactory in Grünheide and can be reached easily and quickly via the A13 (Dresden - Berlin). In addition, the region offers a sparsely populated rural area whose traffic volume is rather low, so that the resulting traffic associated with the implementation of the initiative should not be a strong additional burden on residents and the environment.



The low traffic volume and the proximity to the highway represent a particular locational advantage of the region around the Lausitzring for developing concepts for traffic and transport of tomorrow.

- Proximity to the highway: Experts currently believe that autonomous driving concepts will initially be used primarily on highways. These have no oncoming traffic, are generally in good condition, and lane markings are clearly visible and intact. In addition, there are only motorized vehicles on the highway. All these factors make the highway a good test area.
- The rural area with its low traffic volume also offers the potential to test and further develop more degrees of freedom for autonomous driving on rural and municipal roads. In this sense, it is planned to develop the range and autonomy of the vehicles to be tested step by step up to autonomous driving without persons present in the next few years.

Lusatia - the development laboratory for mobility of the future

When you consider that monthly costs for new cars add up to the following sums:

Micro Car	approx. 330 to 380 € (incl. approx. 117 to 139 € depreciation)
Small Car	approx. 325 to 420 € (incl. approx. 78 to 145 € depreciation)
Compact Car	approx. 330 to 500 € (incl. approx. 86 to 230 € depreciation)
Mid-size Car	approx. 540 to 665 € (incl. approx. 280 to 375 € depreciation)
Upper mid-size Car	approx. 645 to 990 € (incl. approx. 340 to 500 € depreciation)
Large Cars	approx. 1.260 to 1.610 € (incl. approx. 720 to 1.010 € depreciation)

Even if these are all estimated values that autoscout24 gives on its homepage, it still shows how high the monthly or annual sums of money for cars that households spend are. In view of poverty risks and the global crises outlined above, the question arises as to whether mobility cannot be made possible in a different way and at a lower cost for the individual. Autonomously driving and shared vehicles can provide flexible, reliable and safe mobility service both in urban agglomerations, as car sharing already does today, and in rural areas. Mobility as a Service will then be a readily usable mobility option not only for city dwellers in Cologne, Berlin, Munich or Hamburg. Autonomous and self-driving cars also create opportunities for rural areas to do without one, often even two cars of their own and still be mobile, participate in public life and save money.